

Matthew Woody

matthewwoody.github.io • woodyma@sonoma.edu • linkedin.com/mwoodyy

EDUCATION

Sonoma State University

B.S. Applied Mathematics | GPA: 3.41 / 4.00

Expected May 2027

Rohnert Park, CA

- *Relevant Coursework:* Partial Differential Equations, Modern Physics, Numerical Analysis, Mathematical Modeling, Linear Algebra, Data Structures, Probability Theory

Santa Rosa Junior College

A.S. Mathematics & Natural Sciences

May 2025

Santa Rosa, CA

RESEARCH EXPERIENCE

Brookhaven National Laboratory

Atmospheric Science Intern

June 2026 – August 2026

Upton, NY

Advisor: Dr. Mindy Deng

- Integrated NEXRAD Level II polarimetric radar data with minute-resolution CoURAGE surface observations to construct an event-based dataset of 18 cold-front passages across heterogeneous terrain and land-cover regimes
- Developed a Python-based tracking and feature-extraction workflow using polarimetric quality masks, connected-component filtering, lag-correlation propagation estimates, and DBSCAN clustering
- Retrieved and composited drop size and rainfall properties to investigate variation with terrain, season, and Chesapeake Bay breeze occurrence, producing inspectable time-range diagrams and spatial profiles
- Identified a recurring mountain-region enhancement in cold front precipitation, characterized by longer duration, broader occurrence of high instantaneous rain rates, and approximately 2-4 $\times$ greater accumulation than urban region precipitation during the analyzed spring and summer cases
- Project Report Link

University of California, Davis

Undergraduate Atmospheric Science Researcher

October 2025 – Present

Advisor: Dr. Matt Igel

- Built inventory of variations of 7D interacting DoNUT cloud circulations
- Used linear, polynomial and random forest regression to identify broad tendencies between parameterizations

Brookhaven National Laboratory

Atmospheric Science Intern

January 2025 – March 2025

Upton, NY

Advisor: Dr. Mindy Deng

- Applied K-means and DBSCAN clustering to XSACR radar observations from the TRACER campaign to isolate features of Houston's bay breeze
- Developed Python workflows to extract leading-edge position and propagation speed, generate time series statistics and compare frontal velocity with the circulation's overall wind speed

PRESENTATIONS

SSU Annual Math Festival, Rohnert Park, CA, April 2026. "Unsupervised Machine Learning Techniques to Identify Bay Breeze Statistics" (poster).

OTHER EXPERIENCE

SSU Learning and Academic Resource Center
Peer Tutor

August 2026 – Present
Rohnert Park, CA

- *Courses Supported:* Physics I & II, Differential Equations, Calculus, Reasoning and Proof, Mathematical Modeling, Probability Theory, Descriptive Astronomy, Finite Math (in-class tutor)

Santa Rosa Junior College
Finite Math Teaching Assistant

January 2026 – May 2026
Santa Rosa, CA

Brookhaven National Laboratory
Summer Policy Student

May 2025 – August 2025
Upton, NY

City of Petaluma
Water Resources Intern

June 2024 – August 2024
Petaluma, CA

PROFESSIONAL ORGANIZATIONS

American Meteorological Society (2025 – Present)

TECHNICAL SKILLS

Languages: Python, Octave, C++

Libraries: scikit-learn, Xarray, Matplotlib, NumPy, pandas, NetCDF

Tools: Jupyter Notebook, Excel, $\text{\LaTeX}$